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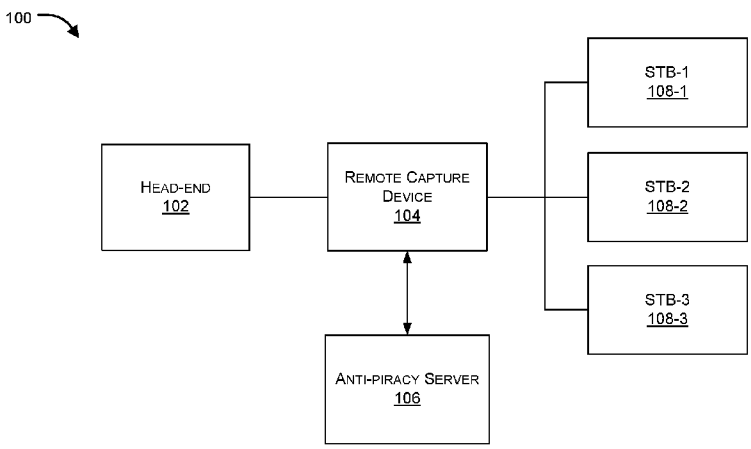

FIG. 1A

(57) Abstract: Embodiments of the present disclosure provide systems and methods for detecting unauthorized distribution of content over a media distribution network such as over a TV network. An embodiment of the present disclosure provides a remote capture device (RCD) that can be configured to tune to a particular channel using a master channel map and a capture channel map, capture content as is being transmitted, extract a watermark from the received content, and compare the watermark with one or more reference watermarks to detect whether the distribution of the content is authorized. The transmission of content can be from any or a combination of head-end, local TV operator and Set Top Box. An anti-piracy server (APS) can also be configured in the proposed system that can be configured to monitor and manage a plurality of RCDs and determine unauthorized distribution of content.

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SYSTEM TO DETECT UNAUTHORIZED DISTRIBUTION OF MEDIA CONTENT

FIELD OF DISCLOSURE

[0001] The present disclosure generally relates to systems and methods for detecting unauthorized distribution of media content. In particular, the present disclosure pertains to methods and systems for detecting unauthorized distribution of media content by a head-end and/or content distributor over a TV network.

BACKGROUND OF THE DISCLOSURE

[0002] The background description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0003] Content distribution, also commonly referred to as TV broadcasting, industry has come a long way, both in terms of its customer reach, and the advancement of technology involved in delivering high end content to users over the user equipments. In a typical Pay-TV scenario, a head-end, generally managed by TV service providers, collects content from media houses that produce multimedia content, encrypts the content, and broadcasts the encrypted content, which can only be received and decrypted by user devices that have decryption key, obtained on subscription of particular content. Television (TV) providers distribute video streams to subscribers by way of implementing conditional access (CA) systems where the CA systems distribute video streams from a head-end of the cable TV provider to a user equipment such as to a set-top box (STB), an IP enabled user device, among others like devices. The head-end includes a hardware that receives video streams generated by the media houses or content producers and distributes such content to the set top boxes within the CA system, which allows access of content only to such STB users who have paid the subscription charge appropriately. Only selected set top boxes are allowed to decode certain video streams according to entitlement information received by them from the cable TV provider and/or head-end.

[0004] Both analog as well as digital networks exist for distribution of such content by Pay-TV service providers, where different sets of CA systems are employed to control distribution of content only to subscribers who have paid for the service. In different business models, a STB user and/or content consumer has the opportunity to select only limited number of channels out of a plurality of channels that are being offered by the service provider. For instance, a STB user can select subscription to content relating to a particular genre, or a particular media house, or only to Hindi channels, Sports channels, News etc., from hundreds of channels being broadcasted by the service provider. Based on such chosen subscription, the head-end side application of the CA system sends entitlement control message (ECM) that can include the required key for decrypting and decoding the content being broadcasted by the head-end side application. The user side application, also referred to as STB side application of the CA can only decrypt channels or content for which the user has paid. Some service providers also offer content on demand where a user can subscribe for only a particular content and not for entire package or channel.

[0005] A Pay-TV service provider broadcasts encrypted content to all set top boxes but only a subset of those set top boxes are given access to specific video programs. For example, only those set top boxes that have ordered a pay per view content are allowed to view the content even though every set top box may receive the content. Once a subscriber orders the pay per view program/content, an entitlement message is broadcasted in encrypted form to all set top boxes, and only specific set top boxes that are entitled to view the content can decrypt it. Inside the decrypted entitlement message is a key that will decrypt the pay per view program, using which the set top box can decrypt the pay per view program as it is received in real-time.

[0006] Pay-TV industry ecosystem is concerned about unauthorized usage of content, which directly harms the interest of the content producers. Content producer suspect unauthorized broadcast of content by the service provider and unauthorized access of the content by the end user. As those skilled in the art can appreciate, set top boxes incorporate elaborate security mechanisms to thwart the efforts of pirates. However, these security mechanisms are occasionally circumvented by pirates or unauthorized users who hack the set top boxes. There exist several solutions for controlling unauthorized access of the content by the end user, while there are very limited solutions for determining whether the head-end side application is broadcasting or offering the content that are legally acquired by the service provider from the content producer.

[0007] Generally, Pay-TV service provider obtains license from the content producer for distribution of content at particular channels for a limited amount of time and for a specific date/time. Pay-TV service provider is to then distribute the content only to the extent as permissible or authorized by the content producer. Distribution of protected content by Pay-TV service provider without authorization of the content owner can harm the interest of content owner and/or content producer. Even if a Pay-TV service provider has a license for distribution of particular content, also referred as program, the distribution should be limited for the time and schedule as approved by the content owner. Repeated distribution/broadcast, primetime broadcast of the content without proper compensation to the content owner/producer can also harm the interest of the content owner/producer. It is therefore desirable that distribution of content from the head-end side to the STB is monitored to determine distribution of unauthorized content.

[0008] There also exists a problem of time-bound evidence collection to prove unauthorized distribution of content. It has been observed that even if the content producers or enforcement agencies suspect unauthorized distribution of content, they have very limited options to collect evidence and prove the matter in court. Unauthorized distribution and/or consumption of content should therefore be detectable, for which legally enforceable evidence needs to be gathered to protect the interest of the content producer.

[0009] Therefore, there exists a need for systems and methods for detecting unauthorized distribution of content and collecting evidence of any violation of interest of the content owner/producer.

[0010] All publications herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

[0011] In some embodiments, the numbers expressing quantities or dimensions of items, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term “about.” Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

[0012] As used in the description herein and throughout the claims that follow, the meaning of "a," "an," and "the" includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of "in" includes "in" and "on" unless the context clearly dictates otherwise.

[0013] The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. "such as") provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

[0014] Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

OBJECTS OF THE INVENTION

[0015] An object of the present disclosure is to provide a method and system for monitoring and detecting unauthorized distribution of content over a TV network.

[0016] Another object of the present disclosure is to provide a method and system for collecting evidence of unauthorized distribution of protected content over a TV network.

[0017] Another object of the present disclosure is to detect unauthorized distribution of protected content and to create evidence of unauthori007Aed content distribution over an analog and/or digital TV network.

[0018] Another object of the present disclosure is to provide a system and method for detecting and deterring TV signal piracy in analog and digital broadcast distribution network.

[0019] Another object of the present disclosure is to provide a method and system for determining unauthorized distribution of content by any service provider at an anti-piracy agency.

[0020] Another object of the present disclosure is to provide a remote capture device (RCD) that is configured to monitor and detect unauthorized distribution of content by a head-end system.

[0021] Another object of the present disclosure is to provide an anti-piracy server for monitoring and detecting unauthorized distribution of content by any head-end system within a particular geographical area.

[0022] Another object of the present disclosure is to determine location of the piracy in real-time, along with information on rough distribution network or head-end that is distributing the unauthorized content.

[0023] Another object of the present disclosure is to provide a piracy intelligence network that can collaboratively detect and collect legally enforceable evidence of unauthorized content distribution.

[0024] Another object of the present disclosure is to detect suspected unauthorized use of content and dispatch a quick response team to location of unauthorized usage to verify and apprehend pirates.

SUMMARY OF THE INVENTION

[0025] The present invention generally relates to systems and methods for detecting unauthorized distribution of media content (interchangeably termed as content herein). In particular, the present disclosure pertains to methods and systems for detecting unauthorized distribution of media content by a head-end and/or content distributor over a TV network.

[0026] An aspect of the present disclosure provides a system to detect unauthorized distribution of content, the system comprising: an image capture and watermark analysis module configured, at a computing device, to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels being transmitted and to extract a watermark from the captured content; a reference watermark determination module configured, at the computing device, to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising a master channel map (MCM) and a capture channel map (CCM), wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels; wherein the image capture and watermark module is further configured to compare the extracted watermark with the reference watermark to determine any or combination of the content being aired at a wrong time; mismatch between the channel ID and the frequency; and the unauthorised distribution of said content.

[0027] In an embodiment, the channel ID is determined based on continuous scanning of the plurality of channels being transmitted and image or pattern recognition of each channel of the plurality of channels. In an embodiment, any or a combination of the reference watermark determination module and the image capture and watermark analysis module is configured on any or a combination of a remote capture device and an anti-piracy server. In an embodiment, the image capture and watermark analysis module is further configured to determine any or a combination of absence of the watermark, the reference watermark being masked, the reference watermark being modified and the reference watermark being obliterated.

[0028] In an embodiment, the system is further configured to perform, based on determination of unauthorised distribution of said content, any or a combination of actions selected from a group consisting of: (a) block further distribution of content, at least in part, from a head-end, a local cable operator or a set top box that performed unauthorised distribution of the content; (b) determine location of the head-end that performed unauthorised distribution of the content; (c) determine location of the local cable operator that performed unauthorised distribution of content; (d) determine location of the set top box that performed unauthorised distribution of the content; and (e) report said locations to any or a combination of content owner, law agency and quick response team. In an embodiment, the system is further coupled with a TV signal matching system to measure TV audience and derive TRPs there from.

[0029] Another aspect of the present disclosure relates to a method to detect unauthorised distribution of content, said method comprising the steps of: configuring, at a computing device, an image capture and watermark analysis module to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels and to extract a watermark from the captured content; configuring, at a computing device, a reference watermark determination module to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising a master channel map (MCM) and a capture channel map (CCM), wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels; capturing, at said image capture and watermark analysis module, the content transmitted on said particular channel; extracting, at said image capture and watermark analysis module, the watermark from the captured content; retrieving, at said reference watermark determination module, the reference watermark based on the particular time and the particular channel comparing, at said image capture and watermark analysis module, the extracted watermark with the reference watermark to determine any or a combination of (a) the content being aired at a wrong time; (b) mismatch between the channel ID and the frequency; and (c) the unauthorised distribution of said content.

[0030] Various objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like features.

BRIEF DESCRIPTION OF DRAWINGS

[0031] The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0032] FIGs. 1A and 1B illustrates an exemplary schematic diagram of a content distribution network having a remote capture device (RCD) and an anti-piracy server (APS) attached with it for detection and prevention of unauthorized distribution of content in accordance with an embodiment of the present disclosure.

[0033] FIG. 2 illustrates an exemplary network of content distribution where plurality of RCDs, each associated with a head-end, operated in conjunction with the APS in accordance with an embodiment of the present disclosure.

[0034] FIG. 3 illustrates an exemplary module diagram of a remote capture device 300 in accordance with an embodiment of the present disclosure.

[0035] FIG. 4 illustrates an exemplary module diagram of an anti-piracy server 400 in accordance with an embodiment of the present disclosure.

[0036] FIG. 5 illustrates another exemplary network of content distribution showing content owner being operatively coupled with an APS, which in turn is operatively coupled with a RCDs in accordance with an embodiment of the present disclosure.

[0037] FIG. 6 illustrates flow diagram of a method for detecting the unauthorized distribution of content in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0038] In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present invention. It will be apparent to one skilled in the art that embodiments of the present invention may be practiced without some of these specific details.

[0039] Embodiments of the present invention include various steps, which will be described below. The steps may be performed by hardware components or may be embodied in machine-executable instructions, which may be used to cause a general-purpose or special-purpose processor programmed with the instructions to perform the steps. Alternatively, steps may be performed by a combination of hardware, software, and firmware and/or by human operators.

[0040] Embodiments of the present invention may be provided as a computer program product, which may include a machine-readable storage medium tangibly embodying thereon instructions, which may be used to program a computer (or other electronic devices) to perform a process. The machine-readable medium may include, but is not limited to, fixed (hard) drives, magnetic tape, floppy diskettes, optical disks, compact disc read-only memories (CD-ROMs), and magneto-optical disks, semiconductor memories, such as ROMs, PROMs, random access memories (RAMs), programmable read-only memories (PROMs), erasable PROMs (EPROMs), electrically erasable PROMs (EEPROMs), flash memory, magnetic or optical cards, or other type of media/machine-readable medium suitable for storing electronic instructions (e.g., computer programming code, such as software or firmware).

[0041] Various methods described herein may be practiced by combining one or more machine-readable storage media containing the code according to the present invention with appropriate standard computer hardware to execute the code contained therein. An apparatus for practicing various embodiments of the present invention may involve one or more computers (or one or more processors within a single computer) and storage systems containing or having network access to computer program(s) coded in accordance with various methods described herein, and the method steps of the invention could be accomplished by modules, routines, subroutines, or subparts of a computer program product.

[0042] If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0043] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

[0044] Exemplary embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. These embodiments are provided so that this disclosure will be thorough and complete and will fully convey the scope of the invention to those of ordinary skill in the art. Moreover, all statements herein reciting embodiments of the invention, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future (i.e., any elements developed that perform the same function, regardless of structure).

[0045] Thus, for example, it will be appreciated by those of ordinary skill in the art that the diagrams, schematics, illustrations, and the like represent conceptual views or processes illustrating systems and methods embodying this invention. The functions of the various elements shown in the figures may be provided through the use of dedicated hardware as well as hardware capable of executing associated software. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the entity implementing this invention. Those of ordinary skill in the art further understand that the exemplary hardware, software, processes, methods, and/or operating systems described herein are for illustrative purposes and, thus, are not intended to be limited to any particular named element.

[0046] Each of the appended claims defines a separate invention, which for infringement purposes is recognized as including equivalents to the various elements or limitations specified in the claims. Depending on the context, all references below to the "invention" may in some cases refer to certain specific embodiments only. In other cases it will be recognized that references to the "invention" will refer to subject matter recited in one or more, but not necessarily all, of the claims.

[0047] All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

[0048] Various terms as used herein are shown below. To the extent a term used in a claim is not defined below, it should be given the broadest definition persons in the pertinent art have given that term as reflected in printed publications and issued patents at the time of filing.

[0049] The term “watermark” as used herein means an image, text or pattern or any combination of a singular or plurality of these that is embedded in content being transmitted.

[0050] The present invention generally relates to systems and methods for detecting unauthorized distribution of media content (interchangeably termed as content herein). In particular, the present disclosure pertains to methods and systems for detecting unauthorized distribution of media content by a head-end and/or content distributor over a TV network.

[0051] The present invention generally relates to systems and methods for detecting unauthorized distribution of media content (interchangeably termed as content herein). In particular, the present disclosure pertains to methods and systems for detecting unauthorized distribution of media content by a head-end and/or content distributor over a TV network.

[0052] An aspect of the present disclosure provides a system to detect unauthorized distribution of content, the system comprising: an image capture and watermark analysis module configured, at a computing device, to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels being transmitted and to extract a watermark from the captured content; a reference watermark determination module configured, at the computing device, to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising a master channel map (MCM) and a capture channel map (CCM), wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels; wherein the image capture and watermark module is further configured to compare the extracted watermark with the reference watermark to determine any or combination of the content being aired at a wrong time; mismatch between the channel ID and the frequency; and the unauthorised distribution of said content.

[0053] In an embodiment, the channel ID is determined based on continuous scanning of the plurality of channels being transmitted and image or pattern recognition of each channel of the plurality of channels. In an embodiment, any or a combination of the reference watermark determination module and the image capture and watermark analysis module is configured on any or a combination of a remote capture device and an anti-piracy server. In an embodiment, the image capture and watermark analysis module is further configured to determine any or a combination of absence of the watermark, the reference watermark being masked, the reference watermark being modified and the reference watermark being obliterated.

[0054] In an embodiment, the system is further configured to perform, based on determination of unauthorised distribution of said content, any or a combination of actions selected from a group consisting of: (a) block further distribution of content, at least in part, from a head-end, a local cable operator or a set top box that performed unauthorised distribution of the content; (b) determine location of the head-end that performed unauthorised distribution of the content; (c) determine location of the local cable operator that performed unauthorised distribution of content; (d) determine location of the set top box that performed unauthorised distribution of the content; and (e) report said locations to any or a combination of content owner, law agency and quick response team. In an embodiment, the system is further coupled with a TV signal matching system to measure TV audience and derive TRPs therefrom.

[0055] Another aspect of the present disclosure relates to a method to detect unauthorised distribution of content, said method comprising the steps of: configuring, at a computing device, an image capture and watermark analysis module to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels and to extract a watermark from the captured content; configuring, at a computing device, a reference watermark determination module to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising a master channel map (MCM) and a capture channel map (CCM), wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels; capturing, at said image capture and watermark analysis module, the content transmitted on said particular channel; extracting, at said image capture and watermark analysis module, the watermark from the captured content; retrieving, at said reference watermark determination module, the reference watermark based on the particular time and the particular channel comparing, at said image capture and watermark analysis module, the extracted watermark with the reference watermark to determine any or a combination of (a) the content being aired at a wrong time; (b) mismatch between the channel ID and the frequency; and (c) the unauthorised distribution of said content.

[0056] FIGs. 1A and 1B illustrate exemplary schematic diagrams of a content distribution network having a remote capture device 104 and anti-piracy server 106 attached with it for detection and prevention of unauthorized distribution of content, in accordance with an embodiment of the present disclosure. As shown in FIG. 1A and FIG. 1B, a head-end 102 is configured to distribute content, preferably encrypted content, to one or more set top boxes

(STBs), such as STB 108-1, STB 108-2 and STB 108-3, collectively and interchangeably referred to as STB 108 hereinafter.

[0057] In an aspect, STB 108 can be any user device that has modules of conditional access system to receive and decrypt content that the user has subscribed. In an exemplary implementation, STB 108 can be any user device including but not limiting to IP TVs, home gateways, mobile devices, and desktop computer devices, which can receive encrypted content from head-end 102, decrypt the received encrypted content using keys received with the entitlement control message, and output the decrypted content over one or more audio/video devices.

[0058] In an exemplary implementation, head-end 102 can be any server device that can receive raw content, encrypt the raw content using modules of conditional access system (CAS) and distribute the encrypted content to the STBs 108. Generally a service provider gets a license from content producer and/or content owner to distribute content through head-end 102 to one or more STBs 108 that it is operatively coupled with. The content producer and/or law enforcement agency may be interested in monitoring the content being distributed by the head-end 102 to its consumer STBs 108 to ensure that only legally authorized content is being distributed. In an exemplary implementation, distribution medium and/or network can be a cable TV network, or an IP TV network or a direct to home satellite based network for distributing encrypted content by the head-end.

[0059] In an embodiment, the proposed system/architecture can include a remote capture device 104 (or a computing device), also interchangeably referred to as RCD 104 hereinafter, that can be attached with/in the distribution network and can be configured to capture samples of content being distributed by the head-end 102 to STBs 108, and help determine unauthorized distribution of content. In an exemplary implementation, as illustrated in FIG. 1A, RCD 104 of the present disclosure can be configured to receive analog signal(s) being transmitted by the head-end 102, and can be configured to tune itself to receive a particular channel and/or particular program, capture the encrypted signal, decrypt the encrypted signal and extract a watermark from the decrypted signal so as to determine whether the content being transmitting is authorized or not. In another exemplary implementation, as illustrated in FIG. 1B, one or more RCD 104 of the present disclosure can be operatively coupled to one or more STB 108 that receive digital signal(s) being transmitted by the head-end 102, and can be tuned to receive a particular channel and/or particular program, capture the encrypted signal, decrypt the encrypted signal and extract a watermark from the decrypted signal so as to determine whether the content being transmitting is authorized or not. In an exemplary implementation, RCD 104 of the present disclosure can be configured to analyze extracted watermark, and determine if the content is authorized or not.

[0060] In an exemplary implementation, RCD 104 of the present disclosure can have details about content that is authorized to be distributed by the head-end 102 at a scheduled time, based on which if it is detected that the head-end 102 (managed by a service provider) distributes/distributed any content that was not authorized by the content producer for distribution by the respective service provider, the RCD 104 can detect such transmission and help deter such unauthorized distribution of content.

[0061] In another exemplary implementation, RCD 104 of the present disclosure can be configured to tune itself to receive a particular channel and/or program, capture the encrypted signal, decrypt the signal, extract a watermark, and then send the extracted watermark along with channel and/or program details to an anti-piracy server 106 (also interchangeably referred to as APS 106 hereinafter). In an aspect, APS 106 can be configured to perform analysis on the received watermark and evaluate the content to figure out whether distribution of the content in question was authorized/ allowed by the content owner to the service provider (associated with the head-end 102 from where watermark has been captured).

[0062] In an aspect, in order to comply with the requirement of the present disclosure, content being transmitting by the head-end 102 can have an embedded watermark that can be used by the RCD 104 and/or APS 06 to determine association of the content with any content producer. One should appreciate that although parts of the disclosure describe the evaluation of watermark being done at the APS 106, all such analysis can be performed at the RCD 104 itself, or by any other configured device/server/computing unit, all of which are completely within the scope of the present disclosure. It is also to be appreciated that RCD and/or APS and/or any other device that is configured to interpret and evaluate content being transmitted by the head-end 102 can be wirelessly or through a wired connection be connected with the head-end 102, and all such implementations are completely within the scope of the present disclosure.

[0063] In an exemplary implementation, RCD 104 can be a wireless device that can receive encrypted content and/or other essential details such as a master channel map (MCM) that contains the mapping of frequencies and the channel/program identifier, a capture channel map (CCM) that contains channel ID/program ID and the timestamp (schedule) information and decryption keys. In different implementations, RCD 102 can be a smart autonomous device that can operate as a standalone system or can be combined with the head-end 102. In example implementation, RCD 104 can be configured to download the MCM and CCM from the APS 106. In example implementation, the RCD 104 can also download a electronic program guide (EPG) data, which include program schedule, channel identifier, program identifier and the like from head-end 102.

[0064] In an exemplary implementation, RCD 104 of the present disclosure can be configured to capture content being transmitting by the head-end 102 by periodically tuning to different channels. RCD 104 can be configured to decrypt the captured encrypted signal using appropriate entitlement keys, and can extract watermark embedded with the content. In an aspect, a content producer/owner can incorporate one or more watermarks with the content so as to trace its unauthorized use. As the RCD 104 extracts the watermark from the content, it can identify association of content with particular content producer, and verify if the content was authorized for distribution by the head-end 102. In another exemplary implementation, RCD 104 can be configured to extract and capture the content, decrypt it, extract the watermark and transmit the watermark along with the timestamp, program ID, channel ID, among other details, as configured by the proposed system to the APS 106. In an exemplary implementation, RCD 104 can create a log of identified content along with the timestamp and can further use these details for determining authorized distribution of content by the head-end 102.

[0065] In an aspect, APS 106 can receive watermark and other details from RCD 104, and analyze them to determine authorized distribution of content. In an exemplary implementation, APS 106 can check whether the content associated with the extracted watermark was authorized for distribution by the head-end 102 at the given time. In an exemplary implementation, APS 106 can compare the watermark of the content as captured by the RCD 104 with the watermark of expected content that were authorized for distribution by the head-end 102 at the given time, wherein APS 106 can detect unauthorized distribution if the expected watermark (interchangeably termed as a reference watermark herein) and the extracted watermark do not match.

[0066] FIG. 2 illustrates another exemplary representation of a content distribution network 200 with a plurality of RCDs 204, each associated with a respective head-end 206, and operated in conjunction with an APS 202 in accordance with an embodiment of the present disclosure. As shown in FIG. 2, an anti-piracy server 202, interchangeably referred hereinafter as APS 202, can be configured to detect authorized distribution of content by any of the head-ends such as head-end-1 206-1, head-end-2 206-02,…, to head-end-N 206-N, referred collectively as head-end(s) 206 hereinafter. In another exemplary implementation, each head-end 206 can have at-least one RCD 204 attached with it. For instance, head-end-1 206-1, head-end-2 206-2, and head-end N 206-N, can be respectively operatively coupled with RCD-1 204-1, RCD-2 204-2, and RCD-3 204-3 attached with it for monitoring content being transmitted/distributed. As the head-ends 206 transmits/distribute content to STBs, the content can be captured by the corresponding RCD 204. For instance, when head-end 206-1 transmits content A to any or a combination of STB-1 208-1, STB-2 208-2,…,and STB-X 208-X, the same content A can be captured by RCD-1 204-1.One should appreciate that the proposed arrangement is completely exemplary in nature, and any other implementation can be configured such that, for instance, the RCD's 204 may not even be in physical proximity to the corresponding head-ends 206 and may be configured to wirelessly or through wired means receive content from the respective head-end. Similarly, it may be possible to have one to many mapping between the RCD's 204 and/or head-ends 206 and all such embodiments are completely within the scope of the present disclosure.

[0067] In an aspect, APS 202 can be a server that can be, for instance, maintained by a central anti-piracy agency, which can maintain a list of authorized content and/or schedule for transmission/distribution of authorized content by different head-ends 206. As the same content may be transmitted by different head-ends 206 and at different times by a single head-end, all authorized distribution schedule of broadcast can be maintained by the APS 202. In an exemplary implementation, APS 202 can be configured to install software upgrades on RCDs 204, monitor health status of RCDs 204, provide frequency/location channel mapping to RCDs 204, remotely configure the RCDs 204, and provide firmware updates to the RCDs 204, along with performing any other configured function. One can appreciate that APS 202 and APS 106 of FIG. 1A and FIG. 1B may be the same device, and have been referred with different numbers so as to maintain flow of the description. Similarly other components such as RCDs and STBs of FIG. 1A and FIG. 1B and FIG. 2 may be configured to perform similar functionality.

[0068] In an exemplary implementation, RCD 204 can be placed with head-ends 206 and can be configured to capture content being transmitted by the head-ends 206, extract watermark from the content, and send watermark to the APS 202 for determining authorized distribution of content. In an exemplary implementation, RCD 204 can be placed with head-ends 206 and can be further configured to capture content being transmitted by the head-ends 206, extract the watermark, and at its own end, determine unauthorized distribution of content. Upon detection of unauthorized distribution of content, RCD 204 can report the incident to the APS 202, which can then take further action.

[0069] In an aspect, RCD 204 can capture the signal as being transmitted by the head-end 206 to STBs 208 in real time. In another exemplary implementation, RCD 204 can be configured to tune to different channels at periodic intervals, and capture content being transmitted, and extract mark(s)/watermark(s) embedded with the content. RCD 204 can also send extracted watermark, along with channel ID to the APS 202. In an exemplary implementation, RCD 204 can be tuned to a particular channel to extract watermark(s) from the content as per the instruction of the APS 202, which may be aware of the time and channel ID when the watermark will be present in the content. In an exemplary implementation, RCD 204 can be configured to receive live channel map and information about the time slot to extract/capture the watermark that will be broadcasted on the channel being monitored.

[0070] In an aspect, before RCD 202 can extract the watermark, RCD 202 can tune to and capture channel/information periodically, which maybe once in say two days for instance. RCD 202 can then capture frame of content, which can include the watermark and send that frame of content to the APS 202 for further analysis and identification of unauthorized content distribution. In an exemplary implementation, APS 202 can send a Master Channel Map (MCM), which include a mapping of channel ID and/or program ID with the frequency at which the channel should appear to the RCD 204. MCM can also include information about mapping of program ID with the channel ID and/or frequency. After complete update of the MCM, RCD 204 can be configured to receive a Capture Channel Map (CCM), which include a list of channel IDs and/or program IDs and date-time combination from the APS 202, wherein the CCM can include channel ID and schedule of the programs, which may be different and unique for each and every head-end 206. In an exemplary implementation, RCD 204 can includes a reference watermark determination module (not shown) that can retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables including CCM and MCM. In an exemplary implementation, RCD 204 can tune to a particular channel, capture encrypted signal of that channel from respective head-end 206, and extract watermark present in the frame of content. In an exemplary embodiment, RCD 204 can compare extracted watermark of the tuned channel with the expected watermark, and if a mismatch is found, the RCD 204 can generate a trigger message indicative of unauthorized distribution of content. In an exemplary implementation, extracted watermark can be compared with watermarks of authorized content. In another exemplary implementation, if a match between the extracted watermark and the expected watermark is not found, it may indicate unauthorized distribution of content by the head-end 206. In an alternative embodiment, extracted watermark can be sent to the APS 202, and comparison of extracted watermark with expected watermark can be performed at the APS 202.

[0071] In an aspect, RCD 204 can be programmed to extract/acquire image of the watermark at predetermined times and intervals and match with a master record of authorized watermark and authorized schedule. In an exemplary implementation, RCD 204 may receive the record of authorized watermark and authorized schedule either from the service provider (also referred as broadcaster) or from the APS 202.

[0072] In another aspect, APS 202 can be configured to receive extracted watermark from one or more RCDs 204, and analyze received extracted watermark for determining any mismatch with expected watermarks. APS 202 analyze extracted frame and/or watermark, identify and verify the watermark code violation for determination of unauthorized distribution. In an exemplary implementation, APS 202 can be configured to remotely manage the RCD 204, integrate the watermark code, capture schedule on the fly, and archive all analyzed data for audit. In an exemplary implementation, APS 202 can receive channel ID/program ID information along with captured image from the RCD 204 along with the extracted watermark. In an exemplary implementation, APS 202 can be configured to receive the updated MCM from the RCD 204, which sends the updated MCM based on input of APS 202 and STBs 208. In an exemplary implementation, upon receiving the MCM from the RCD 204, the APS 202 can analyze and identify the channel where watermark code capture is needed, and can create the CCM with the channel ID, date and time for RCD 204 to act upon. In an aspect, MCM and CCM can be used by the RCD 204 to tune to a particular channel at a specific time and collect image/frame of content containing watermark.

[0073] In an exemplary implementation, APS 202 can be configured to monitor health of one or more RCDs 204, restart and reset the RCD's 204, and send live channel map to the RCD's 204. In an exemplary implementation, APS 202 can also send time slots and/or commands for the RCD 204 to switch to different channels and extract associated watermark information. In an aspect, APS 202 can also be configured to manage and monitor the RCDs 204 to regularly send the MCM and CCM to manage and monitor function logs, and version control of the RCDs 204. In an exemplary implementation, APS 202 can be configured to receive, store and archive extracted watermark along with other details. APS 202 can also be configured to manage user access, reporting and analysis of data log and evidence relating to unauthorized distribution of content.

[0074] In an exemplary implementation, APS 202 can generate an audit report on performance of each head-end 206 within the network. APS 202 can also provide secure access to content owners, service providers (broadcaster), law enforcement agencies to view and access performance of one or more head-ends 206.

[0075] In an exemplary implementation, APS 202 can enable determination of location head-end 206 that transmitted the unauthorized content using geo-location of the RCD 204 and/or based on the registered geo-location of the head-end 206. In an exemplary implementation, APS 202 can generate a heat-map based on authorized distribution of content by a head-end 206 along with its geographical location so as to quickly enable the content owner and law enforcement agencies to determine location of head-end 204 that is distributing unauthorized content.

[0076] In an exemplary implementation, APS 202 can be integrated with plurality of quick response teams located at different locations, which upon receiving an alert from the APS 202, can reach at the location of head-end 206 that transmitted the unauthorized content. In an exemplary implementation, APS 202 can guide a quick response team that is nearest to the location of the head-end 206 that transmitted the unauthorized content. In another exemplary implementation, APS 202, upon detection of unauthorized distribution of content, can send a trigger SMS, email and/or website alerts and alarms to a quick response team that is stationed strategically to immediately proceed to the map location and address of the head-end 206.

[0077] In an aspect, upon reaching the premises of the head-end 206, quick response team can collect evidence of broadcast, block transmission of unauthorized content, and file a complaint based on the collected evidence. The quick response team can also transmit evidence in real-time to the APS 202 or to other appropriate/configured law enforcement agencies. Using the teachings of the present disclosure, a response time of less than 24 hours can be achieved for detecting, locating, and locking unauthorized distribution of content. In an exemplary implementation, APS 202 and RCD 204 can have in-built redundancy that can also include redundancy of GPS mapping services, and integration with quick response team.

[0078] In an exemplary implementation, as soon as an event of unauthorized content distribution by any head-end 206 is detected or registered by APS 202, the APS 202 can create a tracking ticket and keep following up on the tracking ticket till a satisfactory result is achieved by the quick response team. APS 202 can also regularly track progress of the ticket, and if the ticket is not resolved within a predefined time limit, a second level of alert message can be raised to the quick response team. In case the ticket is not resolved even after the second alert, an escalation message can be sent to the higher authority of the enforcement agency.

[0079] In an aspect, ecosystem for implementation of the teachings of present disclosure may require a remote capture device (RCD) and an anti-piracy server (APS).

[0080] FIG. 3 illustrates an exemplary module diagram of a remote capture device 300 in accordance with an embodiment of the present disclosure. As shown in FIG. 3, the remote capture device 300 can include a processor, and a memory that includes one or a combination of a channel identifier capture and transmission module 302, a master channel map receive module 304, a capture channel map receive module 306, and an image capture and watermark analysis module 308. In an exemplary implementation, RCD 300 can be configured to capture a signal as is being transmitted by the head-end to one or more STBs in real-time. In an exemplary implementation, RCD can be configured to tune to different channels at periodic intervals, capture content being transmitted, and extract marker/watermark embedded with the content. Channel identifier capture and transmission module 302 can be configured to extract channel identifier(s) of tuned channel(s) and transmit the channel identifier(s) to, for instance, the anti-piracy server (APS). In an exemplary implementation, channel identifier capture and transmission module 302 may be required in case content is being transmitted from the head-end to STB using an analog channel. In an exemplary implementation, RCD can tune to a particular channel to extract watermarked image from the content as per the instruction of the APS, which may be aware of the time and channel ID when the watermark will be present in the content.

[0081] In an exemplary embodiment, module 302 can continuously scan all channels being transmitted by the head-end, can identify the logo on each channel using pattern and/or image recognition techniques and accordingly label each of the channel being transmitted, such labels being considered as channel identifiers that can be transmitted to the APS.

[0082] In an exemplary implementation, master channel map receive module 304 can be configured to enable the RCD to receive a master channel map (MCM) from the APS, wherein the MCM can include a mapping of channel ID or program ID with the frequency at which respective channels/programs are transmitted by the head-end. Such a list may be updated regularly by APS and can be sent to the RCD so that it can also keep the latest MCM for mapping information.

[0083] In an exemplary implementation, capture channel map receive module 306 can be configured to enable the RCD to, upon configuration/validation of the MCM, receive a list of channel identifiers and date-time combinations from the APS in the form of a capture channel map (CCM), which is unique for each and every head-end/distributor and therefore different for each RCD. Such a CCM can be configured to enable reception of a live channel map and information about time slots to extract/capture the watermark that will be broadcasted on the channel being monitored. In an exemplary implementation, the capture channel map receive module 306 can be configured to receive channel map that includes mapping of channel ID and the day-time time schedule.

[0084] In an exemplary implementation, image capture and watermark analysis module 308 can be configured to enable the RCD to capture the watermarked image of the content being transmitted at selected channel ID. Image capture and watermark analysis module 308 can also be configured to analyze the captured watermark, also referred as extracted watermark, to determine unauthorized distribution of content. Image capture and watermark analysis module 308 can further be configured to match the extracted watermark with expected image(s) such that if a match is not found, module 308 can conclude that unauthorized content is being transmitted. Image capture and watermark analysis module 308 can further be configured to extract the watermark as per the time-schedule by the APS using the MCM and CCM, and then send the watermark information to the APS for further analysis. Module 308 can further be configured to extract/acquire watermark at predetermined times and intervals and match with a master record of authorized watermark and authorized schedule. In an exemplary implementation, RCD may receive the record of authorized watermark and authorized schedule, either from the service provider, also referred as broadcaster, or from the APS.

[0085] FIG. 4 illustrates exemplary functional modules of an anti-piracy server 400 in accordance with an embodiment of the present disclosure. APS 400 of the present disclosure can include a channel identifier receive module 402 that can be configured to receive channel identifiers from the RCDs and/or head-ends, a channel identifier analysis based MCM generation module 404 that can be configured to generate a master channel map based on the channel identifier received, a capture channel map generation module 406 that can be configured to generate a channel map of channel IDs with data-time schedule, and a watermark receive and analysis module 408 that can be configured to receive watermark from at least one RCD and analyze the received watermark with reference to the reference watermark.

[0086] In an exemplary implementation, channel identifier receive module 402 can be configured to receive channel identifiers containing a list of channel IDs from different RCDs or head-ends. In an embodiment, RCD can determine channel identifiers and send the details to the APS. In an exemplary implementation, channel identifier analysis based MCM generation module 404 can be configured to enable the APS to generate a master channel map of channel ID (identifier) and frequency of transmission used by the head-ends. Master Channel Map (MCM) can include a mapping of channel identifiers or program identifiers with the frequency at which the channels/programs are transmitted by the head-end, wherein the MCM can upon generation/ updation be transmitted by the APS to the RCD.

[0087] In an exemplary implementation, capture channel map generation module 406 can be configured to generate a Capture Channel Map (CCM) that includes a list of channel IDs with their respective date-time combinations. CCM can include channel ID and schedule of the programs, which may be different and unique for each and every head-end.

[0088] In another exemplary implementation, capture channel map generation module 406 can get inputs from broadcasters regarding what watermark image is to be broadcast at what time and on what channel and create the capture channel map (CCM) accordingly. The CCM can be sent to capture channel map receive module 306.

[0089] In an aspect, watermark receive and analysis module 408 can be configured to receive extracted watermark from a plurality of RCDs, and analyze received extracted watermark for determining any mismatch with the expected/stored reference watermarks. Module 406 can analyze the extracted channel frames and/or watermark, and identify/verify the watermark code violation for determining unauthorized distribution.

[0090] In an exemplary implementation, watermark receive and analysis module 408 can be configured to receive extracted watermarks from one or more RCDs along with channel IDs, and/or RCD identifiers, and match the extracted watermark(s) with expected watermark(s) that were supposed to be transmitted by the head-end over the channel ID in question.

[0091] In an aspect of the present disclosure, a network of wireless RCDs can be configured in strategic locations, for instance, in each cable TV head-end and can be configured to capture, analyze, and transmit watermark data to a central server system, for instance the APS. In another aspect, RCDs can be programmed to acquire the watermarks at predetermined times and intervals and match them with a master record/reference watermarks, and schedule provided by the broadcaster.

[0092] In another aspect, the Anti-Piracy Server (APS) at a command center can receive multiple watermarks from thousands of field located RCDs, analyze them for a mismatch of the expected watermarks. The APS can also remotely manage the RCDs, integrate the watermark code and capture schedules on the fly and archive all analyzed data for audit. In another aspect, the APS can provide secure access to users from broadcaster, providers, and others to view and export data on the performance of each head-end. In yet another aspect, integration with mapping and GIS servers can also be done to allow images of heat maps, identifying piracy event locations down to street address level (as against head-ends) and guide mobile QRTs on the ground in real time. A positive hit of unauthorized distribution of content can trigger an SMS, email and website alerts and alarms, providing a Quick Response Team (QRT) stationed strategically to immediately proceed to map location and address provided by the system. In an aspect, the entire sequence of activity, from analysis to detection, alert and response can be automatically documented to be directly used for quick filing of prosecution cases.

[0093] In an exemplary implementation, RCD can send raw master channel maps to the APS, which can then analyze and identify channels where watermark capture is needed. The APS can then create the Capture Channel Map (CCM) with the channel id, date and time for the RCD to act upon and transmits it back to the RCD. In an implementation, the APS will need to manage, monitor and maintain all the data, records and instructions, including syncing the CCMs and MCMs. Once the CCM is received by the RCD, it captures and transmits the watermark(s) back to the APS, which then identifies and matches the captured /extracted watermarks with its master template from the broadcaster. The APS provides online access to view and alert for any exceptions or piracy detected.

[0094] According to one embodiment, the present disclosure also facilitates detection of pirated channels at the sub-distribution level of a local cable operator. In another aspect, use of a licensed and legal DTH (Direct to Home) set top box to acquire a channel legally but distribute it illegally is also a function of the present disclosure, which allows remote identification and barring of the rogue STB. Furthermore, in another aspect, detection of content where the fingerprint/watermark has been masked, modified, and/or obliterated is also possible with the present disclosure.

[0095] FIG. 5 illustrates another exemplary network 500 of content distribution showing content owner 502 being operatively coupled with an APS 504, which in turn is operatively coupled with a RCD 506 in accordance with an embodiment of the present disclosure. As shown, content owner 502 can be communicatively coupled with an APS 504, and can either through APS 504, or directly be configured to transmit scan timings 508 along with channel details/identifiers to the RCD 506. Based on such data provided by the content owner 502/APS 504, RCD 506 can be configured to receive content from head-end 510 (intended for set-top box 512) and extract watermark/fingerprints from the broadcasted content.

[0096] In an aspect, RCD 506 of the present disclosure can include a tuner/channel scanner 514 that can be configured to receive channel-timing mapping information from either the APS 504 and/or from the content owner 502. RCD 506 can further include a channel screen capture module 516 configured to capture channel information/content broadcasted by head-end 510. RCD 506 can further include an automatic channel recognition engine 518 that can be configured to determine the channel to which the content pertains and extract fingerprint/watermark information from the content. RCD 506 can further include a geo-tag module 520 configured to geo-tag head-end from which it has received information/content and send such location details while reporting back to the APS 504 and/or directly to the content owner 502.

[0097] In an aspect, data received from the RCD 506 can be collected in a data collection center 522 of the APS 504, which can accordingly update the master channel map (MCM) 534 that can be stored in or be operatively accessible to the APS 504. APS 504 can further be configured to match the received fingerprint/watermark information with reference images/watermarks in association with the content owner 502 (say at block 526 of content owner 502) or individually, and report the findings to the content owner 502. Content owner 502 can, in an instance, process the data/comparison output or perform any other data analytics at 528 and in case distribution of unauthorized content is detected/determined, a response team 530 can be alerted along with reporting/tracking/managing such issues through a web-based/application interface 532.

[0098] In an aspect, system of the present disclosure can identify a channel map or channel line-up in the case of DTH, Cable, Terrestial RF TV (legacy Doordarshan), among other Digital and Analog transmission and distribution networks, including IP. This fulfills a great need because position of a channel in the serial order is a key part of its viewership appeal and premium placement is expensive. In many cases, cable operators surreptitiously shift the location of prime channels in favor of other, 'more paying' ones. The present disclosure can also perform TV Audience measurement and derive TRPs or identification of "Prime time" based on viewership measurement, including measuring who is watching in the sample or panel home. For this function, the proposed system can be coupled with TV signal matching technology.

[0099] FIG. 6 illustrates an exemplary flow diagram 600 of a method for detecting the unauthorized distribution of content in accordance with an embodiment of the present disclosure. As shown in FIG. 6, method of the present disclosure can include, at step 602, receiving at the RCD, updated MCM and CCM from an APS. At step 604, the method can include receiving at the RCD, signal being transmitted by a corresponding head-end, and at step 606, tuning the RCD to a selected channel at a defined time/date. At step 608, the method can include the step of extracting at the RCD, a watermark, which at step 610, can be matched with expected/reference watermark/image or can be transmitted to the APS for such matching as shown at step 612. At step 614, based on the outcome of the matching, unauthorized distribution of content, if any, can be detected.

[00100] As used herein, and unless the context dictates otherwise, the term “coupled to” is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms “coupled to” and “coupled with” are used synonymously. Within the context of this document terms “coupled to” and “coupled with” are also used euphemistically to mean “communicatively coupled with” over a network, where two or more devices are able to exchange data with each other over the network, possibly via one or more intermediary device.

[00101] It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refers to at least one of something selected from the group consisting of A, B, C ….and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc. The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the appended claims.

[00102] While embodiments of the present disclosure have been illustrated and described, it will be clear that the disclosure is not limited to these embodiments only. Numerous modifications, changes, variations, substitutions, and equivalents will be apparent to those skilled in the art, without departing from the spirit and scope of the disclosure, as described in the claims.

ADVANTAGES OF THE INVENTION

[00103] The present disclosure provides a method and system for monitoring and detecting unauthorized distribution of content over a TV network.

[00104] The present disclosure provides a method and system for collecting evidence of unauthorized distribution of protected content over a TV network.

[00105] The present disclosure detects authorized distribution of protected content and creates evidence of unauthorized content over an analog and/or digital TV network.

[00106] The present disclosure provides system and method for detecting and deterring TV signal piracy in analog and digital broadcast distribution networks.

[00107] The present disclosure provides method and system for determining authorized distribution of content by any service provider by means of an anti-piracy agency.

[00108] The present disclosure provides a remote capture device (RCD) that is configured to monitor and detect unauthorized distribution of content from a head-end system.

[00109] The present disclosure provides an anti-piracy server for monitoring and detecting unauthorized distribution of content by any head-end system within a particular geographical area.

[00110] The present disclosure determines location of the piracy use in real-time, along with determination of rough distribution network or head-end that is responsible for distributing the unauthorized content.

[00111] The present disclosure provides a piracy intelligence network that can collaboratively can detect and collect legally enforceable evidence of unauthorized content distribution.

[00112] The present disclosure detects suspected unauthorized use of content and enables dispatch of a quick response team to location of unauthorized usage to verify and apprehend pirates.

I Claim:

1) A system to detect unauthorized distribution of content, the system comprising:

an image capture and watermark analysis module configured, at a computing device, to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels being transmitted and to extract a watermark from the captured content;

a reference watermark determination module configured, at the computing device, to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising

a master channel map (MCM) and a capture channel map (CCM),

wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels;

wherein the image capture and watermark module is further configured to compare the extracted watermark with the reference watermark to determine any or combination of

the content being aired at a wrong time;

mismatch between the channel ID and the frequency; and

the unauthorised distribution of said content.

2) The system of claim 1, wherein the channel ID is determined based on continuous scanning of the plurality of channels being transmitted and image or pattern recognition of each channel of the plurality of channels.

3) The system of claim 1, wherein any or a combination of the reference watermark determination module and the image capture and watermark analysis module is configured on any or a combination of a remote capture device and an anti-piracy server.

4) The system of claim 1, wherein the image capture and watermark analysis module is further configured to determine any or a combination of absence of the watermark, the reference watermark being masked, the reference watermark being modified and the reference watermark being obliterated.

5) The system of claim 1, wherein the system is further configured to perform, based on determination of unauthorised distribution of said content, any or a combination of actions selected from a group consisting of:

block further distribution of content, at least in part, from a head-end, a local cable operator or a set top box that performed unauthorised distribution of the content;

determine location of the head-end that performed unauthorised distribution of the content;

determine location of the local cable operator that performed unauthorised distribution of content;

determine location of the set top box that performed unauthorised distribution of the content; and

report said locations to any or a combination of content owner, law agency and quick response team.

6) The system of claim 1, wherein the system is further coupled with a TV signal matching system to measure TV audience and derive TRPs there from.

7) A method to detect unauthorised distribution of content, said method comprising the steps of:

configuring, at a computing device, an image capture and watermark analysis module to capture at a particular time, a content transmitted on a particular channel selected from a plurality of channels and to extract a watermark from the captured content;

configuring, at a computing device, a reference watermark determination module to retrieve a reference watermark based on the particular time and the particular channel, wherein the reference watermark is retrieved from mapping tables comprising

a master channel map (MCM) and a capture channel map (CCM),

wherein the MCM comprises mapping of frequency and corresponding channel ID for each channel of the plurality of channels being transmitted and wherein, the CCM comprises mapping of channel ID, program ID and date-time correlation for each channel of the plurality of channels;

capturing, at said image capture and watermark analysis module, the content transmitted on said particular channel;

extracting, at said image capture and watermark analysis module, the watermark from the captured content;

retrieving, at said reference watermark determination module, the reference watermark based on the particular time and the particular channel

comparing, at said image capture and watermark analysis module, the extracted watermark with the reference watermark to determine any or a combination of

the content being aired at a wrong time;

mismatch between the channel ID and the frequency; and

the unauthorised distribution of said content.

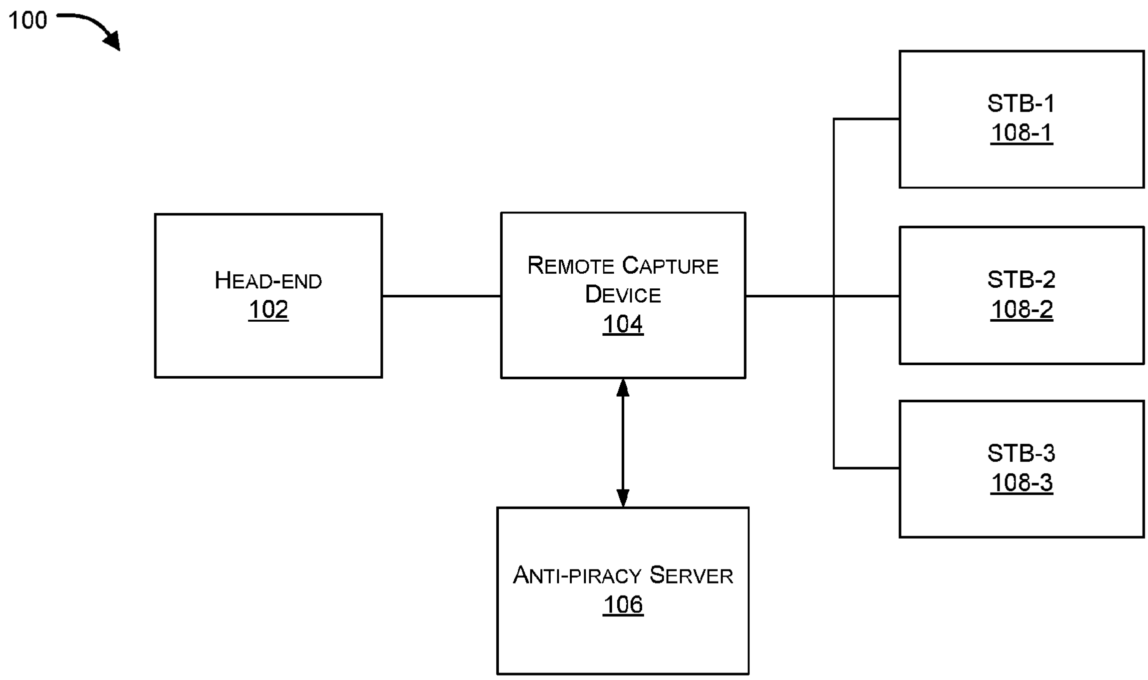
100
HEAD-END
102
REMOTE CAPTURE
DEVICE
104
STB-1
108-1
STB-2
108-2
STB-3
108-3
ANTI-PIRACY SERVER
106

FIG. 1A

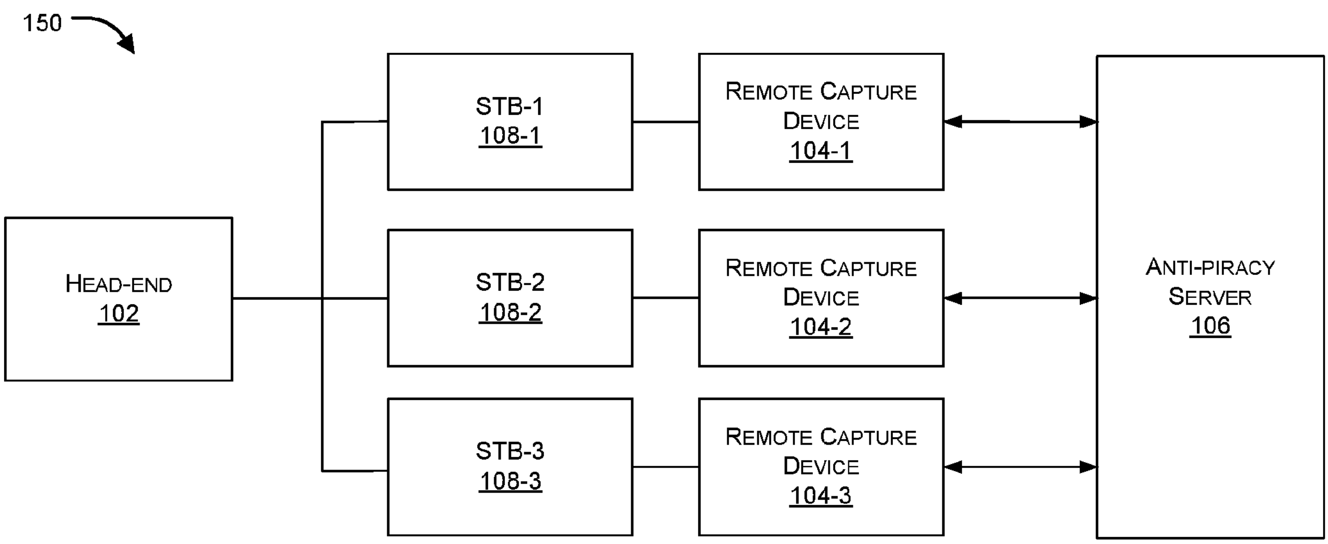
150
HEAD-END
102
STB-1
108-1
STB-2
108-2
STB-3
108-3
REMOTE CAPTURE
DEVICE
104-1
REMOTE CAPTURE
DEVICE
104-2
REMOTE CAPTURE
DEVICE
104-3
ANTI-PIRACY
SERVER
106

FIG. 1B

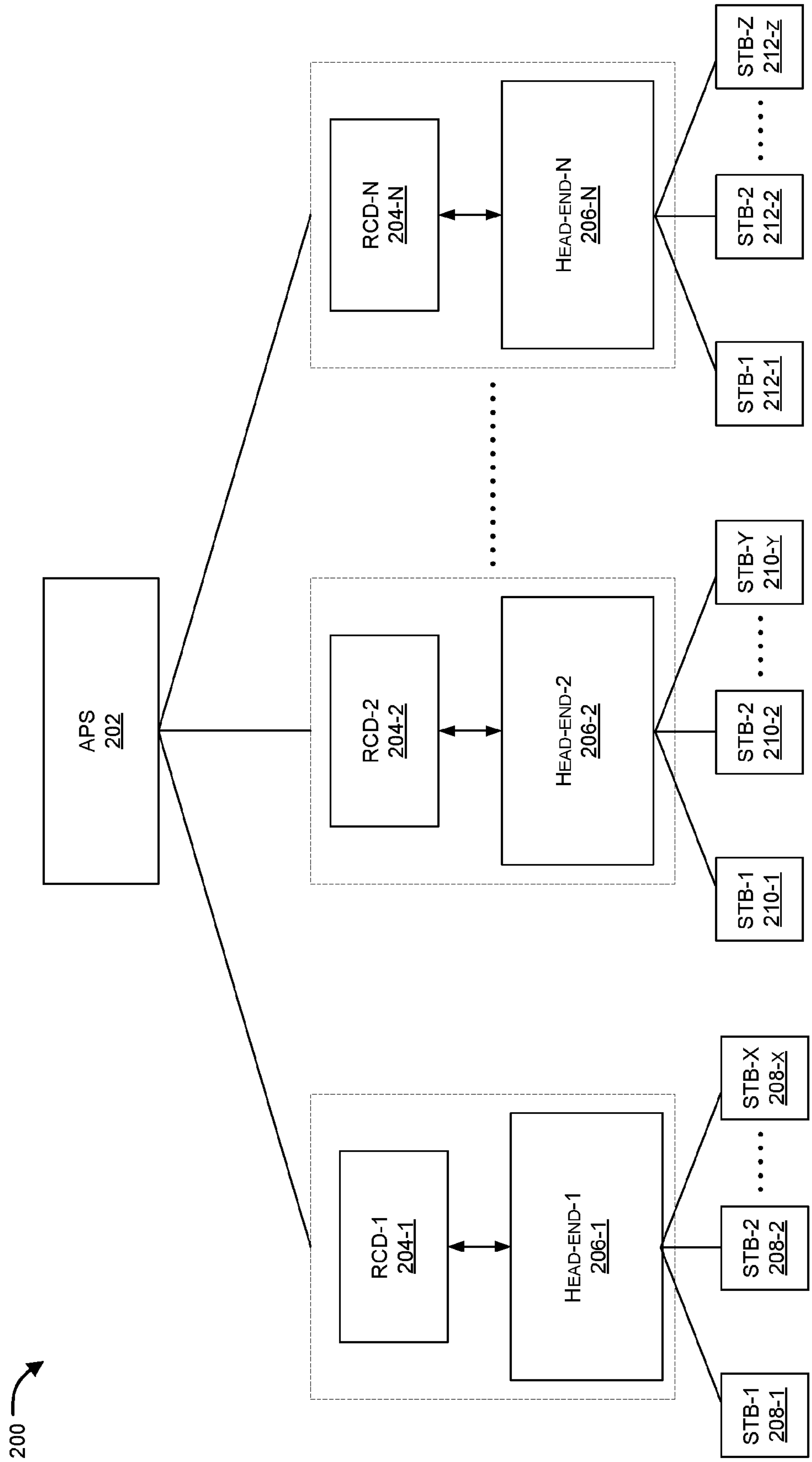
200
APS
202
RCD-1
204-1
HEAD-END-1
206-1
STB-1
208-1
STB-2
208-2
STB-X
208-X
RCD-2
204-2
HEAD-END-2
206-2
STB-1
210-1
STB-2
210-2
STB-Y
210-Y
RCD-N
204-N
HEAD-END-N
206-N
STB-1
212-1
STB-2
212-2
STB-Z
212-Z

FIG. 2

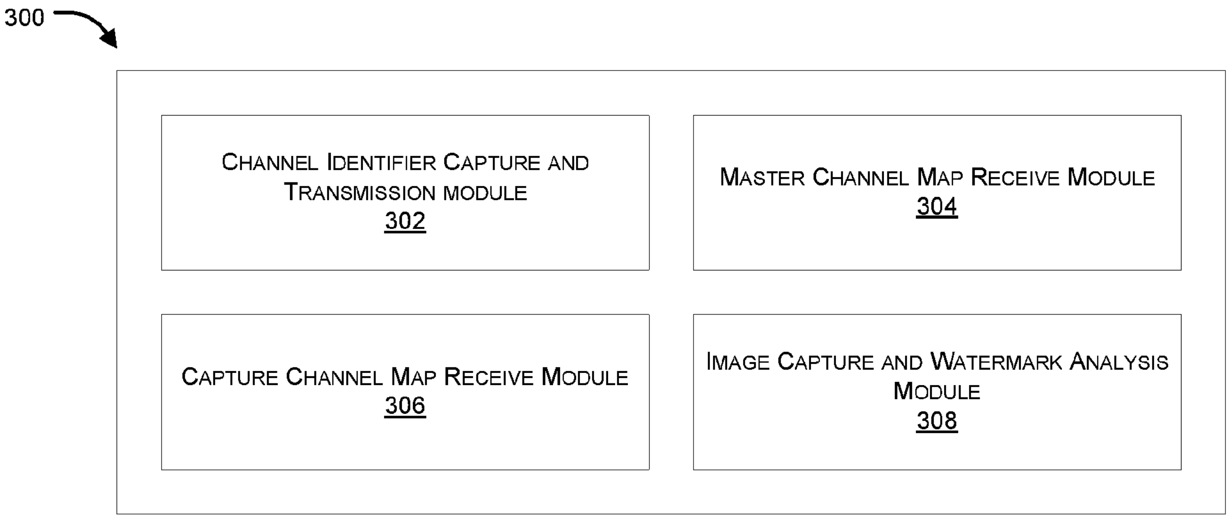
300
CHANNEL IDENTIFIER CAPTURE AND TRANSMISSION MODULE
302
MASTER CHANNEL MAP RECEIVE MODULE
304
CAPTURE CHANNEL MAP RECEIVE MODULE
306
IMAGE CAPTURE AND WATERMARK ANALYSIS MODULE
308

FIG. 3

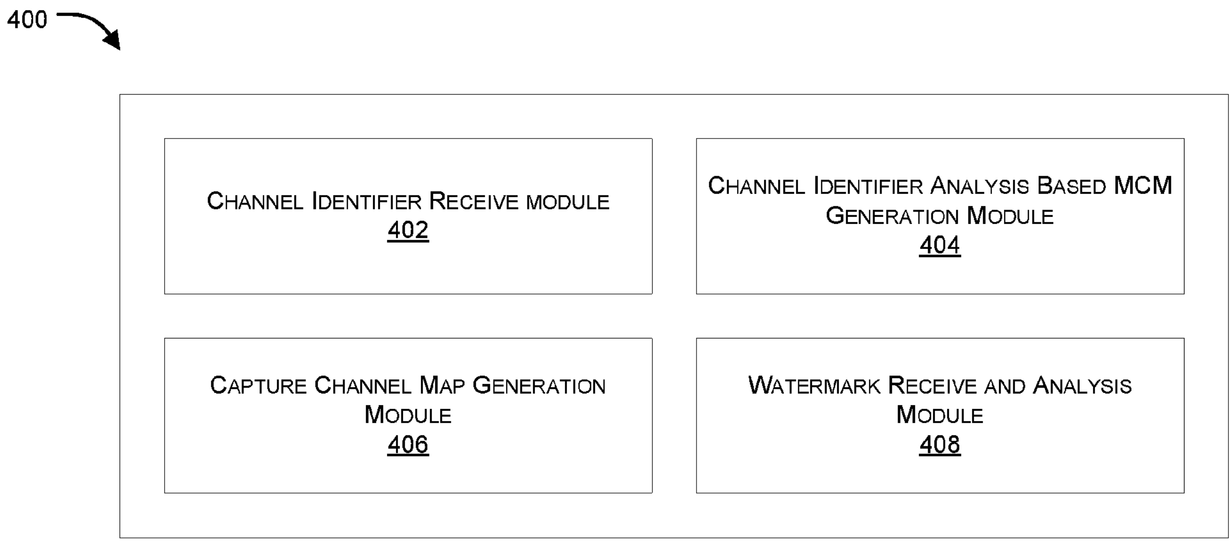
400
CHANNEL IDENTIFIER RECEIVE MODULE
402
CHANNEL IDENTIFIER ANALYSIS BASED MCM GENERATION MODULE
404
CAPTURE CHANNEL MAP GENERATION MODULE
406
WATERMARK RECEIVE AND ANALYSIS MODULE
408

FIG. 4

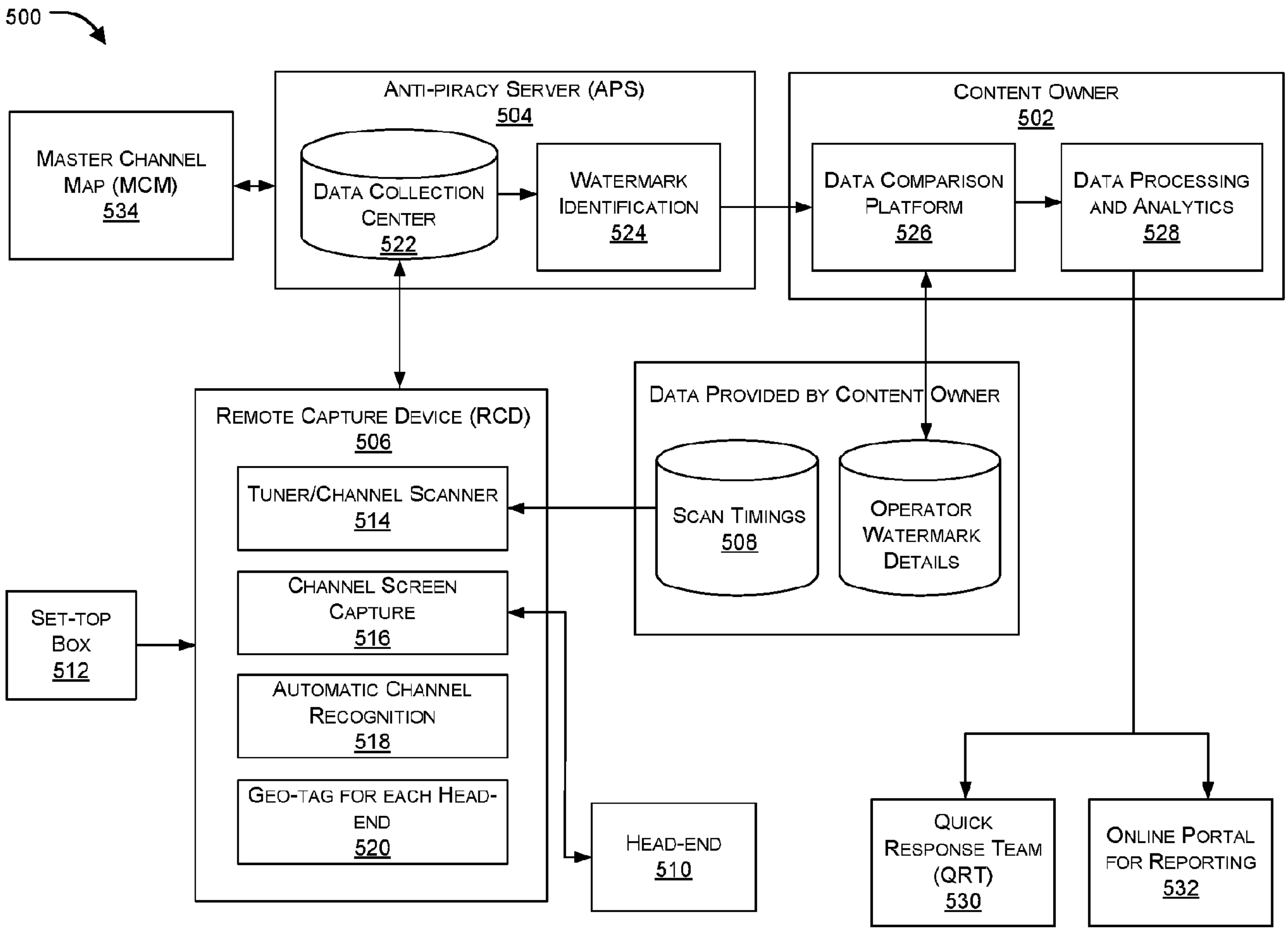
500
ANTI-PIRACY SERVER (APS)
504
CONTENT OWNER
502
MASTER CHANNEL MAP (MCM)
534
DATA COLLECTION CENTER
522
WATERMARK IDENTIFICATION
524
DATA COMPARISON PLATFORM
526
DATA PROCESSING AND ANALYTICS
528
DATA PROVIDED BY CONTENT OWNER
REMOTE CAPTURE DEVICE (RCD)
506
TUNER/CHANNEL SCANNER
514
SCAN TIMINGS
508
OPERATOR WATERMARK DETAILS
CHANNEL SCREEN CAPTURE
516
SET-TOP BOX
512
AUTOMATIC CHANNEL RECOGNITION
518
GEO-TAG FOR EACH HEAD-END
520
HEAD-END
510
QUICK RESPONSE TEAM (QRT)
530
ONLINE PORTAL FOR REPORTING
532

FIG. 5

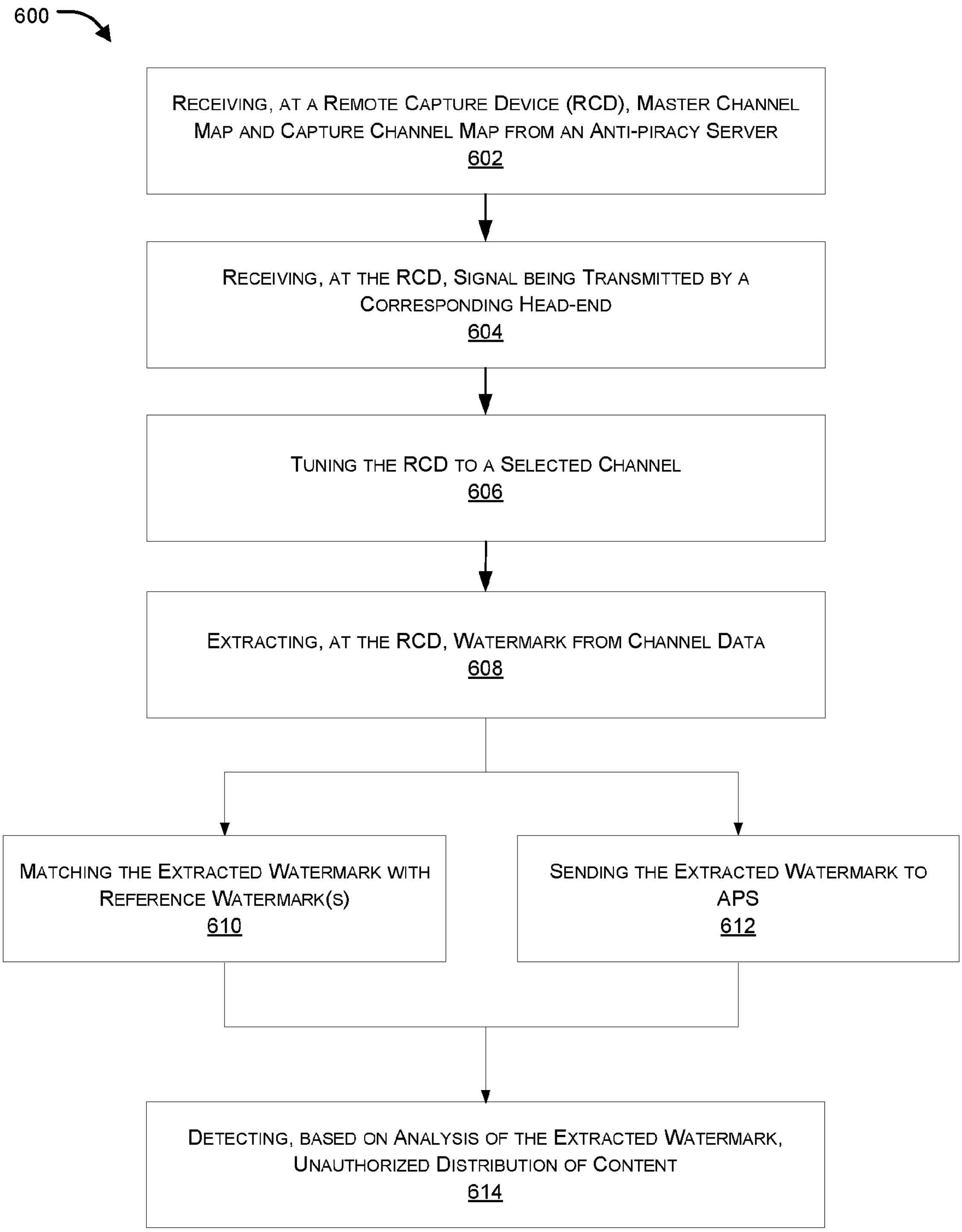
600
Receiving, at a Remote Capture Device (RCD), Master Channel Map and Capture Channel Map from an Anti-piracy Server
602
Receiving, at the RCD, Signal being Transmitted by a Corresponding Head-end
604
Tuning the RCD to a Selected Channel
606
Extracting, at the RCD, Watermark from Channel Data
608
Matching the Extracted Watermark with Reference Watermark(s)
610
Sending the Extracted Watermark to APS
612
Detecting, based on Analysis of the Extracted Watermark, Unauthorized Distribution of Content
614

FIG. 6

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2016/054445

A. CLASSIFICATION OF SUBJECT MATTER
H04N21/00,H04N7/12 Version=2016.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H04N

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

DATABASES: PATSEER, IPO INTERNAL
SEARCH TERMS: WATERMARK, TV, NETWORK, UNAUTHORIZE, CHANNEL, DETECT

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US20150193899 A1 (ANT OZTASKENT, YAROSLAV VOLOVICH) 09 JULY 2015 (09-07-2015) PARAGRAPH [0029], LINES 5-6; PARAGRAPH [0034], LINES 5-9; PARAGRAPH [0047], LINES 4-7; PARAGRAPH [0049], LINE 8; PARAGRAPH [0061], LINES 1-9; PARAGRAPH [0061], LINES 7-13; PARAGRAPH [0064], LINES 7-8; PARAGRAPH [0077], LINES 7-8; PARAGRAPH [0079], LINES 1-4; PARAGRAPH [0100], LINES 1-3	1-7

☐ Further documents are listed in the continuation of Box C.   ☒ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
21-11-2016	21-11-2016

Name and mailing address of the ISA/	Authorized officer
Indian Patent Office Plot No.32, Sector 14,Dwarka,New Delhi-110075 Facsimile No.	Nutishri Lonhare Telephone No. +91-1125300200

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IB2016/054445

Citation	Pub.Date	Family	Pub.Date
US 2015193899 A1	09-07-2015	US 9363519 B2	07-06-2016